

CSE446: Blockchain & Cryptocurrencies

Lecture – 7: Bitcoin - 1

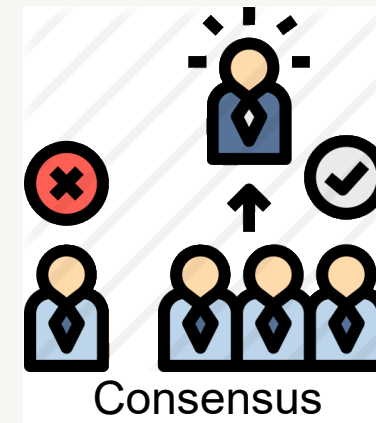
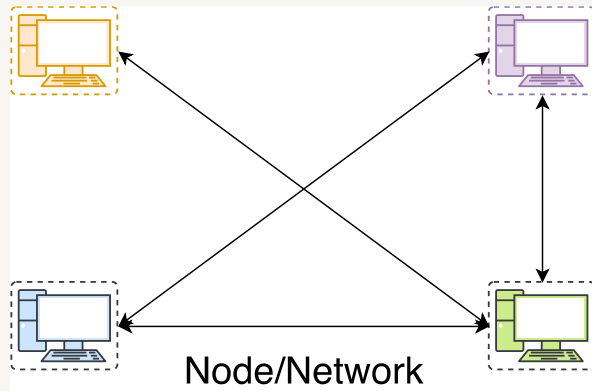
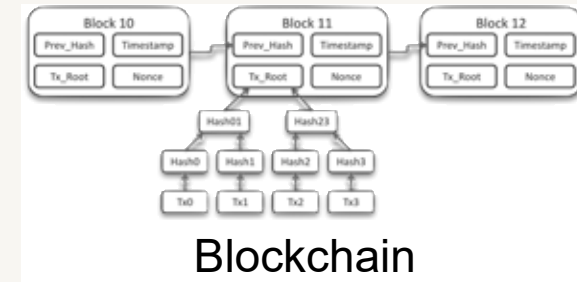
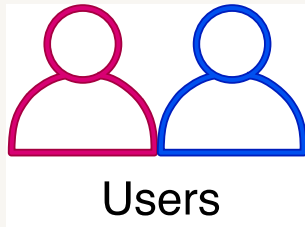


Inspiring Excellence

Agenda

- Bitcoin components
 - Users

Bitcoin/Blockchain components



Decentralised Identity

- Proving the identity of users is a must in many online services
- To create an identity, you need to register to the Service Provider (SP)
- An identity requires a unique identifier to uniquely identify an entity within the system
 - Username -> unique only within a system
 - Email/mobile phone number are universal identifiers
- These identifiers need to be accompanied by a credential (e.g. a password) to prove the ownership of the identifier
- But all these need to rely on a specific SP
 - For emails, it is the Email provider and so on, if such an SP ceases to exist, all services dependent on the identifiers become vulnerable

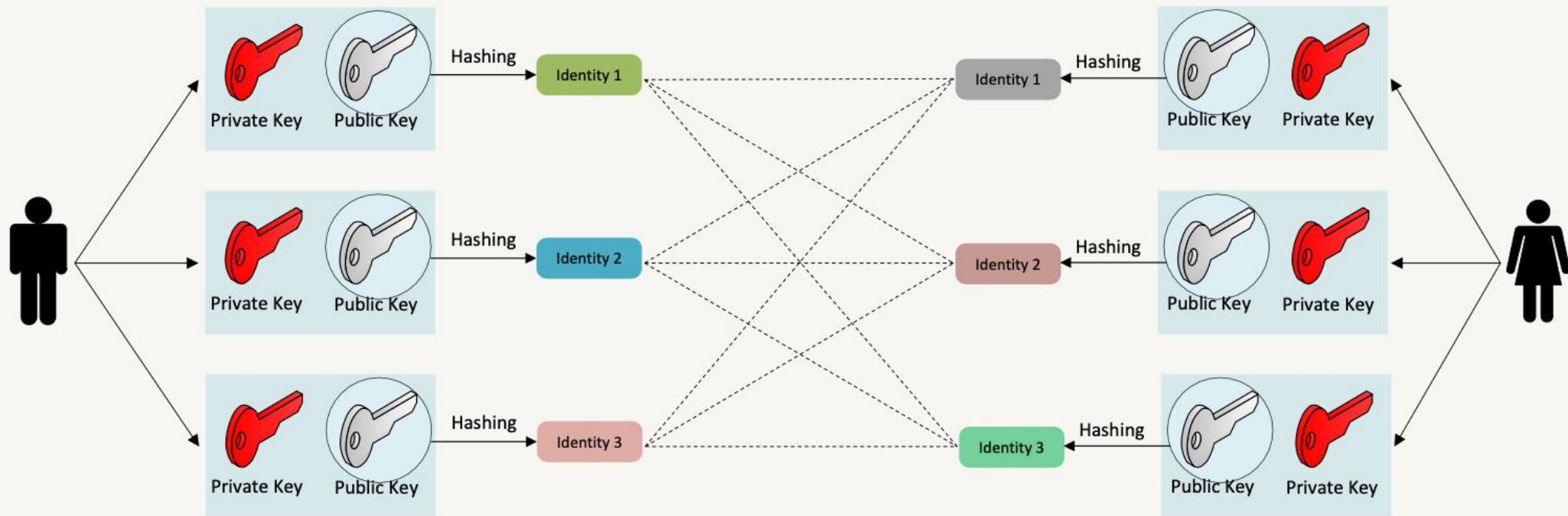
Decentralised Identity

- Decentralised identity is the solution using public key schemes
 - The public key pk acts as an identity
 - The private key sk is the password to prove the ownership of this identity
- This has some advantages:
 - New identities can be generated at will with the *generateKeys* function
 - Also, these new identities cannot be used to uncover your real-world identity, providing a layer of pseudonymous privacy

Decentralised Identity

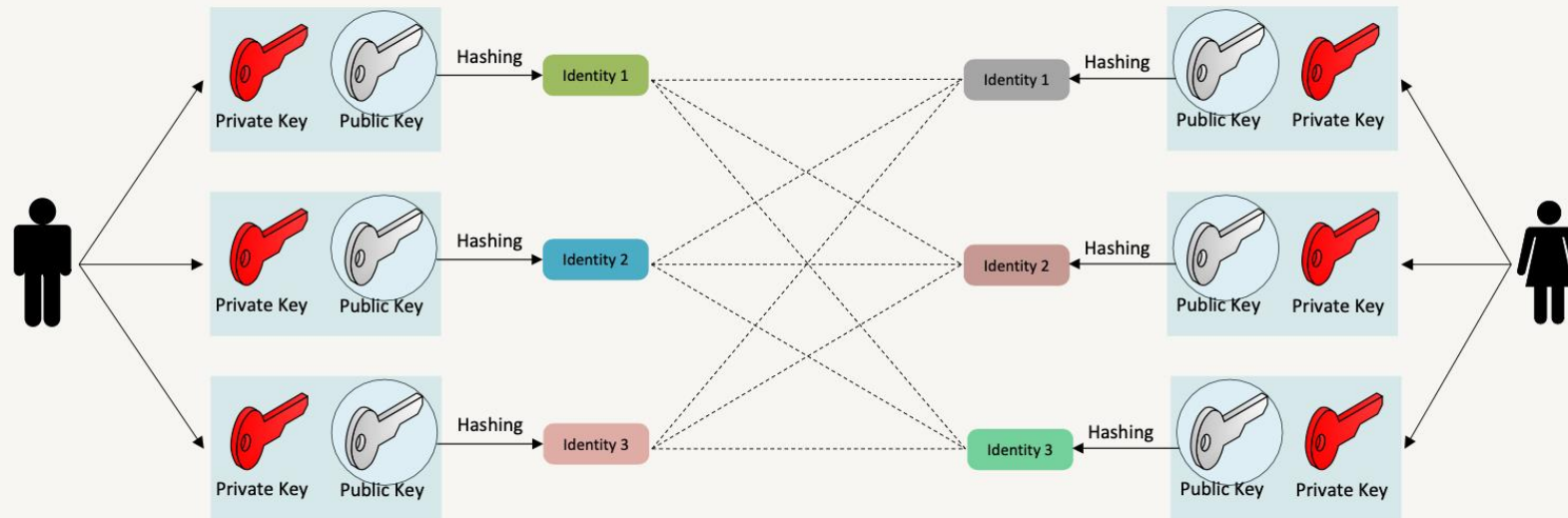
- Public keys are very large
 - You want to hash your public key (pk) to receive an “identity”
- To validate a statement, one must check
 1. if the pk hashes to the identity and
 2. if the message verifies under the public key pk

Decentralised Identity



Almost all (public) blockchain systems adopt this approach

Users



- Public key is used for creating identities
- Such identities can be vetted by a CA (Certificate Authority) for private blockchain systems

Users

- Bitcoin represent users anonymously/pseudonymously
- However, there must be a way to identify a user
- Each user is represented using **an address**, generated by public key cryptography
- Bitcoin uses an elliptic curve (secp256k1) for its public key cryptography
- A user generates a key pair (k_p, k_s)
 - k_p -> represents a public key
 - k_s -> represents a private key
- An **address** can be generated from the public key
- A user **receives** coins with the address
- A user **spends** coins with the private key

Bitcoin address

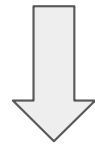
Private key:

5JXesisRRU2Z7HMmwMpNtoiYk1QDMVjV3HLoYMd1PTKEkJhJT1z



Public key:

**045a5f526dfe5d5995bf95f1229e70e21818190883c40ab3590458476ad34aaae5
9bc772b98a587035b452638b59238e2a39e954b43ab7a4f32408664d36ec1575**



Address: **133GT5661q8RuSKrrv8q2Pb4RwSpUTQU1Z**

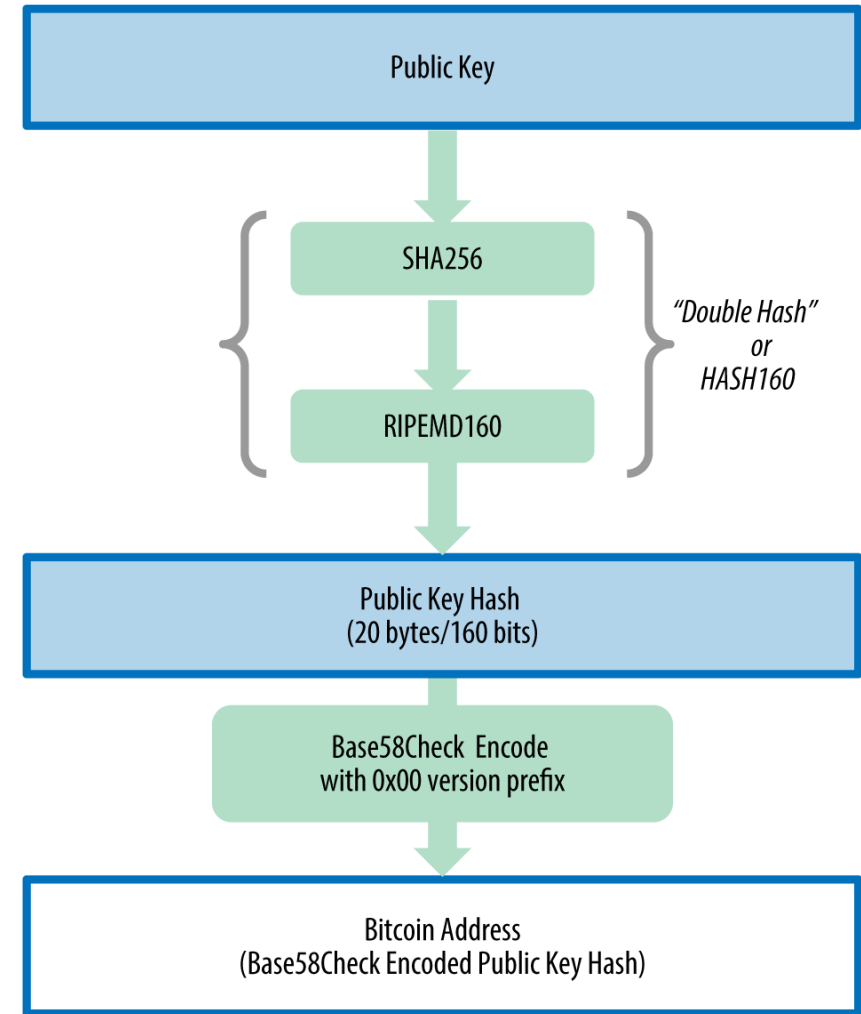
Bitcoin address

Public key:

045a5f526dfe5d5995bf95f1229e70e21818190883c40ab3590458476ad34aaae5
9bc772b98a587035b452638b59238e2a39e954b43ab7a4f32408664d36ec1575

Address: 133GT5661q8RuSKrrv8q2Pb4RwSpUTQU1Z

Public Key to Bitcoin Address



Base64 Encoding

Base64 Table

Value	Char	Value	Char	Value	Char	Value	Char
0	A	16	Q	32	g	48	w
1	B	17	R	33	h	49	x
2	C	18	S	34	i	50	y
3	D	19	T	35	j	51	z
4	E	20	U	36	k	52	0
5	F	21	V	37	l	53	1
6	G	22	W	38	m	54	2
7	H	23	X	39	n	55	3
8	I	24	Y	40	o	56	4
9	J	25	Z	41	p	57	5
10	K	26	a	42	q	58	6
11	L	27	b	43	r	59	7
12	M	28	c	44	s	60	8
13	N	29	d	45	t	61	9
14	O	30	e	46	u	62	+
15	P	31	f	47	v	63	/

- A binary-to-text encoding scheme
 - to represent binary data in an ASCII string format by translating it into a radix-64 representation (radix means the number of unique digits)
- Base64 alphabet:
 - English letters 26 lower + 26 upper + 10 numeral + '+' + '/'

source ASCII (if <128)	M								a								n															
source octets	77 (0x4d)								97 (0x61)								110 (0x6e)															
Bit pattern	0	1	0	0	1	1	0	1	0	1	1	0	0	0	0	1	0	1	1	0	1	1	1	0								
Index	19								22								5								46							
Base64-encoded	T								W								F								u							
encoded octets	84 (0x54)								87 (0x57)								70 (0x46)								117 (0x75)							

Base58 Encoding

Base58 Table

- Base64 alphabets minus six alphabets:
 - 0 (number zero), O (capital o), l (lower L), I (capital i), '+', '/'

Value	Character	Value	Character	Value	Character	Value	Character
0	1	1	2	2	3	3	4
4	5	5	6	6	7	7	8
8	9	9	A	10	B	11	C
12	D	13	E	14	F	15	G
16	H	17	J	18	K	19	L
20	M	21	N	22	P	23	Q
24	R	25	S	26	T	27	U
28	V	29	W	30	X	31	Y
32	Z	33	a	34	b	35	c
36	d	37	e	38	f	39	g
40	h	41	i	42	j	43	k
44	m	45	n	46	o	47	p
48	q	49	r	50	s	51	t
52	u	53	v	54	w	55	x
56	y	57	z				

```
base10 = 123456789

123456789 % 58 = 19
2128565 % 58 = 23
36699 % 58 = 43
632 % 58 = 52
10 % 58 = 10

base58 = [10][52][43][23][19]
base58 = BukQL
```

Encoding
example

Base58 Encoding

Base58 Table

Value	Character	Value	Character	Value	Character	Value	Character
0	1	1	2	2	3	3	4
4	5	5	6	6	7	7	8
8	9	9	A	10	B	11	C
12	D	13	E	14	F	15	G
16	H	17	J	18	K	19	L
20	M	21	N	22	P	23	Q
24	R	25	S	26	T	27	U
28	V	29	W	30	X	31	Y
32	Z	33	a	34	b	35	c
36	d	37	e	38	f	39	g
40	h	41	i	42	j	43	k
44	m	45	n	46	o	47	p
48	q	49	r	50	s	51	t
52	u	53	v	54	w	55	x
56	y	57	z				

- Base-58 encode the word:

C **a** **t**

Character	ASCII dec value				
C	67	$67 * 2^{2*8}$	$67 * 2^{16}$	$67 * 65536$	4390912
a	97	$97 * 2^{1*8}$	$97 * 2^8$	$97 * 256$	24832
t	116	$116 * 2^{0*8}$	$116 * 2^0$	$116 * 1$	116
					4415860

- The word "Cat" in decimal representation: **4415860**

Base58 Encoding

Base58 Table

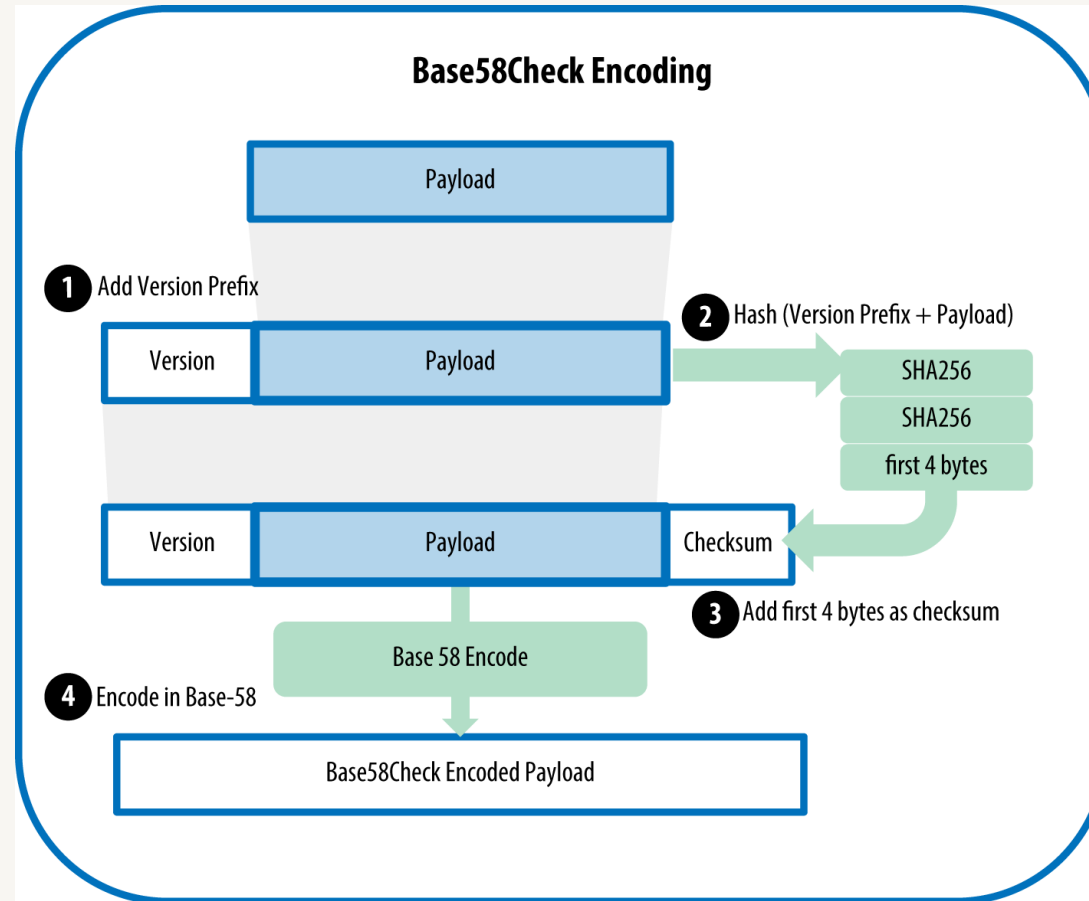
Value	Character	Value	Character	Value	Character	Value	Character
0	1	1	2	2	3	3	4
4	5	5	6	6	7	7	8
8	9	9	A	10	B	11	C
12	D	13	E	14	F	15	G
16	H	17	J	18	K	19	L
20	M	21	N	22	P	23	Q
24	R	25	S	26	T	27	U
28	V	29	W	30	X	31	Y
32	Z	33	a	34	b	35	c
36	d	37	e	38	f	39	g
40	h	41	i	42	j	43	k
44	m	45	n	46	o	47	p
48	q	49	r	50	s	51	t
52	u	53	v	54	w	55	x
56	y	57	z				

- “Cat” decimal representation: **4415860**

Calculate	Equals	Remainder
4415860 / 58	76135	30
76135 / 58	1312	39
1312 / 58	22	36
22 / 58	0	22

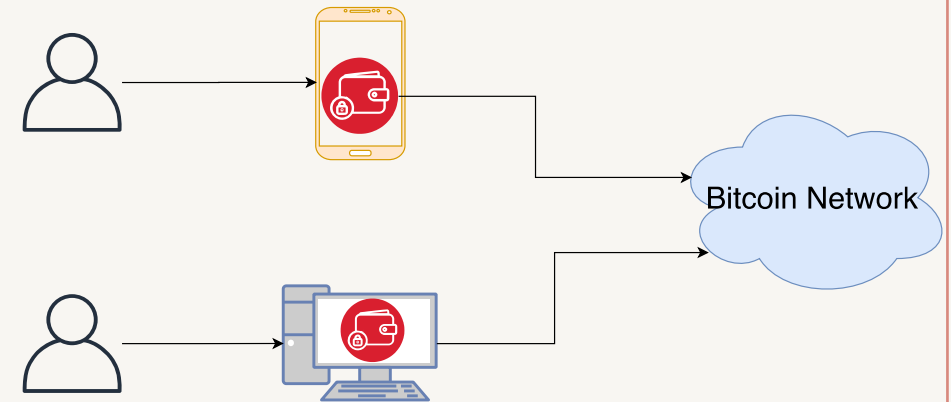
- Remainder values: 22, 36, 39, 30

Bitcoin address



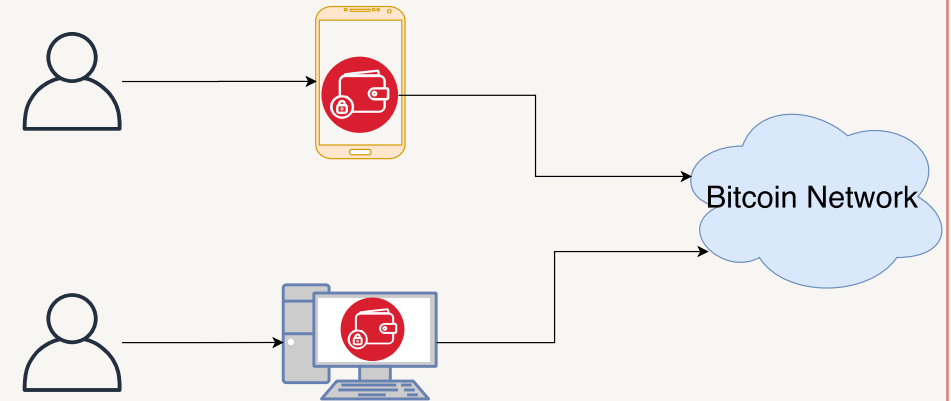
Bitcoin (hot) wallet

- A Bitcoin wallet is a collection of private keys
 - might be used to manage those keys and to make transactions on the Bitcoin network
- It is the entry point for any general users to interact with the bitcoin network
- Can be utilised in a PC, in any smart-device such as a mobile phone or tablet
- Also known as hot wallets as they are always connected to the network
- Examples: Exodus, Electrum, Mycelium



Bitcoin (hot) wallet

- Private keys are kept in encrypted (with a password) formats to ensure their security
- If password is forgotten, there is no way to recover funds attached to that private address
 - unlike other password enabled services, there is no account recovery option
- Strong usability issue
- Advantageous for daily trading or continuous usage
- Less secure (e.g. prone to malware attack)



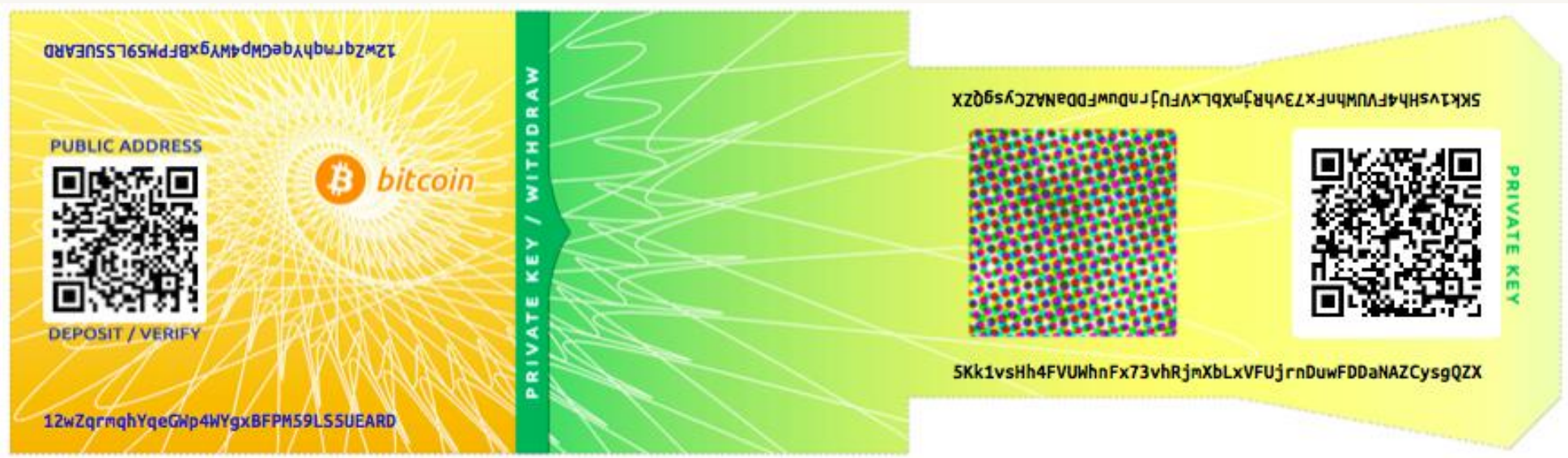
Cold wallet: paper wallet/cold wallet

- Paper wallets are bitcoin private keys printed on paper
 - might include the corresponding bitcoin address for convenience, but this is not necessary because it can be derived from the private key
- They are a very effective way to create backups or offline bitcoin storage, also known as *cold storage/wallet*

Cold wallet: paper wallet/cold wallet

- As a backup mechanism, a paper wallet can provide security
 - against the loss of key due to a computer mishap such as a hard-drive failure, theft, or accidental deletion
 - store it offline in a secret & secure place, even in a bank vault
- As a "cold storage" mechanism, if the paper wallet keys are generated offline
 - never stored on a computer system
- Hence, they are much more secure against hackers, keyloggers, and other online computer threats

Cold wallet: paper wallet/cold wallet



https://raw.githubusercontent.com/bitcoinbook/bitcoinbook/develop/images/mbc2_0410.png

Hardware wallet

- A **hardware wallet** is a special type of bitcoin wallet which stores the user's private keys in a secure hardware device
- They have major advantages over standard software wallets:
 - private keys are often stored in a protected area of a microcontroller, and cannot be transferred out of the device in plaintext
 - immune to computer viruses that steal from software wallets
 - can be used securely and interactively, as opposed to a paper wallet which must be imported to software at some point
 - much of the time, the software is open source, allowing a user to validate the entire operation of the device



<https://www.ledgerwallet.com/images/products/lns/ledger-nano-s-fold-medium.png>



<https://en.bitcoin.it/w/images/en/d/de/Trezor-tx.jpg>

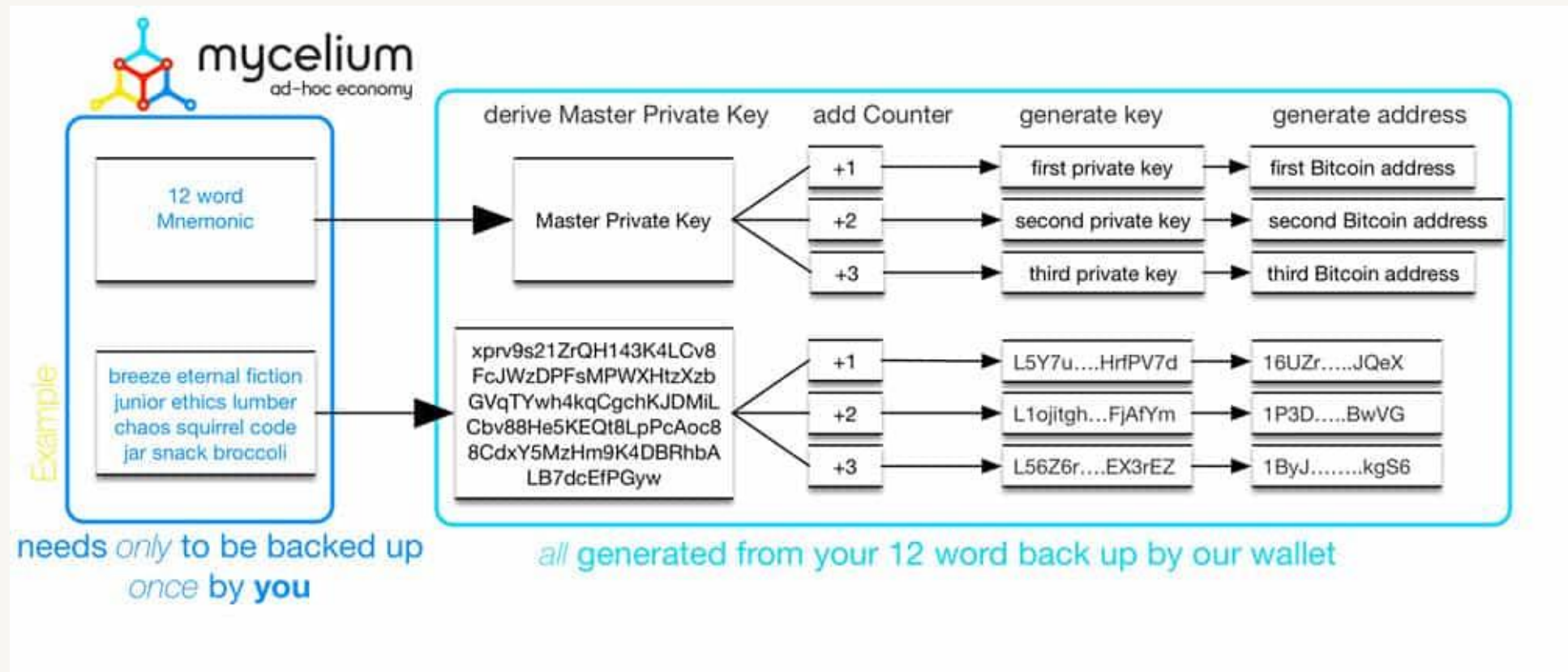
HD wallet

- A Hierarchical Deterministic (HD) is a key creation and transfer protocol which allows creating child keys from a parent key in a hierarchical way
 - Wallets using the HD protocol are called HD wallets
 - The single starting parent key is known as a seed
- The seed allows a user to easily back up and restore a wallet without needing any other information

HD wallet

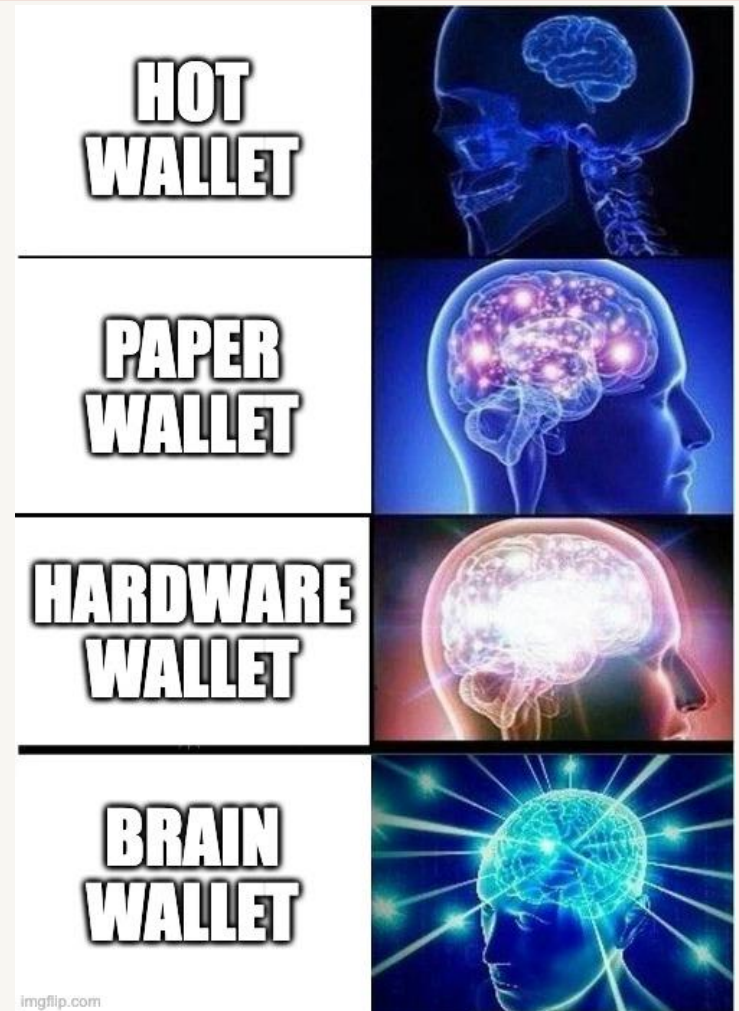
- Seeds are typically serialised into human-readable words in a **Mnemonic** phrase
- A mnemonic phrase, mnemonic recovery phrase or mnemonic seed is a list of words which store all the information needed to recover a Bitcoin wallet
- Such Mnemonic words must be stored securely and must never be typed on any website

HD wallet



Brain wallet

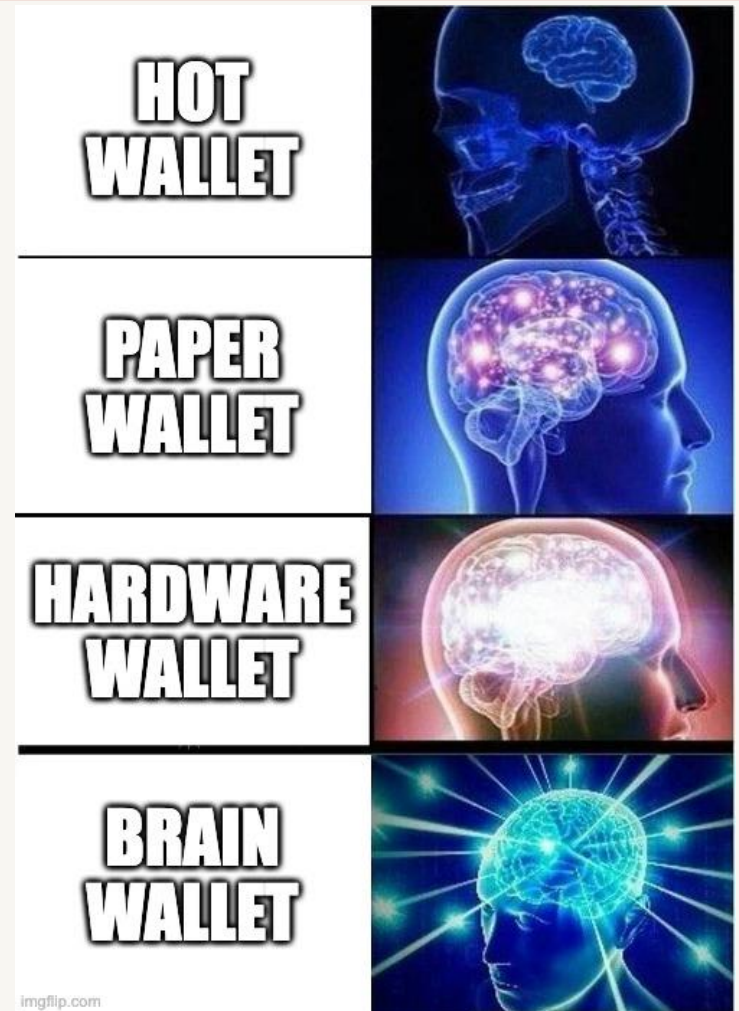
- A brain wallet refers to the concept of storing Bitcoin's private key in one's own mind by memorising a seed phrase
- If the seed is not recorded anywhere, the Bitcoins can be thought of as being held only in the mind of the owner
- If a brain wallet is forgotten or the person dies or is permanently incapacitated, the Bitcoins are lost forever
- Using memory techniques allow them to be memorised and recalled easily



<https://i.imgflip.com/6x41nq.jpg>

Brain wallet

- Private keys will be generated from a seed (mnemonic words/passphrase)
- Use the passphrase to generate a private key
 - E.g. hashing the passphrase
- Then generate the public key from the private key using a standard algorithm
- The passphrase needs to be securely created, otherwise an adversary could easily guess
- If the passphrase is long, it will be difficult to memorise



<https://i.imgflip.com/6x41nq.jpg>

Brain wallet

- To memorise a seed with this method you must invent a story which hits the words as "keynotes"
- Let the key phrases are the following
 - witch collapse practice feed shame open despair creek road again ice least
- "Imagine going through a room and seeing your sister dressed as a **witch**, playing the jenga boardgame until the tower **collapses** and so on"

Brain wallet

"Do you keep your
money in your bank or
at home?"

Me:



In my memories.

